Lesson 38: Signaling, Perfect Bayesian Equilibrium, and Potential Games

AI/ML Mathematics

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1 Motivation and Overview

This lesson covers two topics central to multi-agent RL: *signaling games with perfect Bayesian equilibrium* and *potential games*.

Signaling games extend the Bayesian framework of Lesson 37 to dynamic settings where an informed player communicates through costly actions. The equilibrium concept must combine sequential rationality with belief updating, leading to PBE. The sender's message may fully reveal her type (separating), hide it (pooling), or partially disclose it (semi-separating).

Potential games are games where all incentives are captured by a single scalar function. This guarantees pure NE existence and convergence of simple learning dynamics—a rare property in multi-agent settings.

Signaling and Potential Structure in MARL

In MARL, agents' actions serve dual purposes: achieving rewards and signaling intentions. Understanding PBE predicts what information is communicated in emergent protocols. Potential games provide one of the few classes where independent learners are guaranteed to converge to NE. Potential-based reward shaping leverages this structure to align individual and team objectives.

2 References

- **PBE and signaling:** Fudenberg and Tirole [1991], Chapters 6 and 8; Spence [1973].
- **Potential games:** Monderer and Shapley [1996]; Rosenthal [1973].
- **Multi-agent learning:** Shoham and Leyton-Brown [2008], Ch. 6–7.
- **Extensive-form games:** Osborne and Rubinstein [1994], Ch. 6 and 11.

3 Perfect Bayesian Equilibrium

In dynamic games of incomplete information, BNE (Lesson 37) does not constrain behavior at unreached information sets. We need a refinement combining sequential rationality with belief consistency.

Definition 3.1 (System of Beliefs)**.** A *system of beliefs* μ assigns to each information set h a probability distribution $\mu(\cdot|h)$ over the nodes in h: $\mu(x|h) \geq 0$ for $x \in h$ and $\sum_{x \in h} \mu(x|h) = 1$.

Definition 3.2 (Perfect Bayesian Equilibrium)**.** A *perfect Bayesian equilibrium* (PBE) is a pair (σ, μ) consisting of a behavioral strategy profile σ and a belief system μ satisfying:

(a) **Sequential rationality:** At every information set h of every player i, the strategy σ_i maximizes player i's expected payoff given beliefs $\mu(\cdot|h)$ and the continuation strategies σ_{-i}.

(b) **Belief consistency:** At every information set h reached with positive probability under σ ($\mathbb{P}^\sigma(h) > 0$), beliefs are derived via Bayes' rule:

$$\mu(x|h) = \frac{\mathbb{P}^\sigma(x \text{ is reached})}{\sum_{x' \in h} \mathbb{P}^\sigma(x' \text{ is reached})} \quad \text{for all } x \in h.$$

(c) At information sets reached with zero probability (off the equilibrium path), beliefs $\mu(\cdot|h)$ are unrestricted.

Remark 3.3. The freedom in specifying off-path beliefs is both a strength and a weakness. It allows equilibria to be supported by pessimistic beliefs ("if I see an unexpected message, I assume the worst"), but may permit equilibria sustained by implausible beliefs. Refinements such as the intuitive criterion address this but are beyond our scope.

Proposition 3.4 (Every PBE is a BNE)**.** *Every PBE strategy profile is a Bayesian Nash equilibrium.*

Proof. BNE requires ex ante optimality; PBE requires optimality at every information set (a stronger condition). Optimality at each information set implies optimality of the ex ante weighted average. □

4 Signaling Games

Definition 4.1 (Signaling Game)**.** A *signaling game* has a Sender with private type $\theta \in \Theta$ drawn from prior $p(\theta)$, who chooses message $m \in M$; and a Receiver who observes m, forms belief $\mu(\theta|m)$, and chooses action $a \in A$. Payoffs $u_S(\theta, m, a)$ and $u_R(\theta, m, a)$ depend on all three.

Definition 4.2 (Equilibrium Types)**.** A PBE is *separating* if different types send different messages; *pooling* if all types send the same message; and *semi-separating* if some types pool and others separate.

Example 4.3 (Spence's Job Market Signaling)**.** Worker ability $\theta \in \{L, H\}$ with $\mathbb{P}(H) = \pi$. Worker chooses education $e \geq 0$; employer offers wage $w(e)$. Payoffs: $u_S = w - c(\theta, e)$ and $u_R = \theta - w$, where the *single-crossing property* holds: $c(H, e) < c(L, e)$ for $e > 0$.

Separating PBE. Type H chooses $e_H > 0$, type L chooses $e_L = 0$. Bayes' rule: $\mu(H|e_H) = 1$, $\mu(H|0) = 0$, so $w(e_H) = H$, $w(0) = L$. Incentive compatibility requires:

$$\begin{aligned} H - c(H, e_H) &\geq L \quad \text{(type } H \text{ prefers } e_H\text{)}, \\ L &\geq H - c(L, e_H) \quad \text{(type } L \text{ prefers } e_L = 0\text{)}. \end{aligned}$$

The single-crossing property ($c(L, e) > c(H, e)$) ensures a range of valid e_H exists. Off-path beliefs $\mu(H|e) = 0$ for $e \notin \{0, e_H\}$ support the equilibrium.

Pooling PBE. Both types choose $e^* = 0$: $\mu(H|0) = \pi$, $w(0) = \pi H + (1 - \pi)L$. Off-path: $\mu(H|e) = 0$ for $e > 0$, giving $w(e) = L$. No type deviates since $L - c(\theta, e) < L \leq w(0)$.

Proposition 4.4 (PBE Existence in Finite Signaling Games)**.** *Every finite signaling game has a PBE.*

Proof. Form the agent normal form: one agent per sender-type (choosing m) and one per message (receiver choosing a). By Nash's theorem, this finite game has a mixed NE. Translating back gives σ_S and σ_R; Bayes' rule determines on-path beliefs, and off-path beliefs are chosen freely. Sequential rationality holds because each agent's strategy is a best response. □

5 Potential Games

Definition 5.1 (Exact Potential Game)**.** A finite game $G = (N, (A_i)_{i \in N}, (u_i)_{i \in N})$ is an *exact potential game* if there exists a function $\Phi : A \to \mathbb{R}$ (the *potential function*) such that for every player $i \in N$, every $a_{-i} \in A_{-i}$, and every $a_i, a_i' \in A_i$:

$$u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i}).$$

That is, the change in any player's payoff from a unilateral deviation equals the change in the potential.

Definition 5.2 (Ordinal Potential Game)**.** A finite game G is an *ordinal potential game* if there exists $\Phi : A \to \mathbb{R}$ such that for every i, a_{-i}, and a_i, a_i':

$$u_i(a_i, a_{-i}) > u_i(a_i', a_{-i}) \iff \Phi(a_i, a_{-i}) > \Phi(a_i', a_{-i}).$$

Every exact potential game is ordinal (equal differences have equal signs); the converse is false.

Example 5.3 (Coordination Game as Potential Game)**.** The coordination game with payoffs $(T, L) = (2, 2)$, $(T, R) = (0, 0)$, $(B, L) = (0, 0)$, $(B, R) = (1, 1)$ is an exact potential game with $\Phi(T, L) = 2$, $\Phi(T, R) = 0$, $\Phi(B, L) = 0$, $\Phi(B, R) = 1$. Verification: $u_1(T, L) - u_1(B, L) = 2 = \Phi(T, L) - \Phi(B, L)$; $u_2(T, L) - u_2(T, R) = 2 = \Phi(T, L) - \Phi(T, R)$; and similarly for all other unilateral deviations.

Pure NE in Potential Games

Theorem 5.4 (Potential Games Have Pure NE)**.** *Every finite exact (or ordinal) potential game has a pure Nash equilibrium.*

Proof. Since A is finite, Φ attains its maximum at some a^*. For any player i and deviation a_i': $\Phi(a_i^*, a_{-i}^*) \geq \Phi(a_i', a_{-i}^*)$ by maximality. The potential property gives $u_i(a_i^*, a_{-i}^*) - u_i(a_i', a_{-i}^*) \geq 0$, so a^* is a NE. For ordinal potentials: if $u_i(a_i', a_{-i}^*) > u_i(a_i^*, a_{-i}^*)$ held, the ordinal property would give $\Phi(a_i', a_{-i}^*) > \Phi(a_i^*, a_{-i}^*)$, contradicting maximality. □

6 Convergence of Learning Dynamics

Definition 6.1 (Best-Response Dynamics)**.** In *best-response dynamics*, at each step one player i switches to $a_i^{t+1} \in \arg\max_{a_i} u_i(a_i, a_{-i}^t)$; others keep $a_j^{t+1} = a_j^t$. In *better-response dynamics*, the switch need only strictly improve i's payoff.

Finite Improvement Property

Theorem 6.2 (Finite Improvement Property)**.** *In every finite exact potential game, any sequence of strict unilateral improvements terminates in finitely many steps at a pure NE.*

Proof. If player i switches from a_i^t to a_i^{t+1} with $u_i(a_i^{t+1}, a_{-i}^t) > u_i(a_i^t, a_{-i}^t)$, the potential property gives $\Phi(a^{t+1}) > \Phi(a^t)$. Since A is finite, a strictly increasing sequence of Φ-values must terminate. At termination, no player can improve, so the profile is a NE. □

Corollary 6.3 (Better-Response Convergence)**.** *Better-response dynamics also converge to a pure NE in finitely many steps, since the proof uses only the strict improvement property.*

7 Congestion Games and Rosenthal's Theorem

Definition 7.1 (Congestion Game)**.** A *congestion game* is $(N, E, (A_i), (d_e))$ where N is the player set, E is a finite set of resources, $A_i \subseteq 2^E$ is i's action set (subsets of resources to use), and $d_e : \{1, \ldots, n\} \to \mathbb{R}$ is the delay for resource e depending on congestion $n_e(a) = |\{j : e \in a_j\}|$. Player i's cost is $C_i(a) = \sum_{e \in a_i} d_e(n_e(a))$, with payoff $u_i = -C_i$.

Rosenthal's Theorem

Theorem 7.2 (Congestion Games are Exact Potential Games)**.** *Every congestion game is an exact potential game with* $\Phi(a) = -\sum_{e \in E} \sum_{k=1}^{n_e(a)} d_e(k)$.

Proof. Fix player i, a_{-i}, and two actions a_i, a_i'. Let $n_e = n_e(a_i, a_{-i})$ and $n_e' = n_e(a_i', a_{-i})$. We check three cases for each resource e:

(i) $e \in a_i \cap a_i'$**:** $n_e = n_e'$; contributions cancel in both differences.

(ii) $e \in a_i \setminus a_i'$**:** $n_e = n_e' + 1$. Contribution to $u_i - u_i'$: $-d_e(n_e' + 1)$. Contribution to $\Phi - \Phi'$: $-\sum_{k=1}^{n_e'+1} d_e(k) + \sum_{k=1}^{n_e'} d_e(k) = -d_e(n_e' + 1)$. Equal.

(iii) $e \in a_i' \setminus a_i$**:** $n_e' = n_e + 1$. Contribution to $u_i - u_i'$: $d_e(n_e + 1)$. Contribution to $\Phi - \Phi'$: $d_e(n_e + 1)$. Equal.

Summing over all resources gives $u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a_i', a_{-i})$. □

Corollary 7.3 (Congestion Games Have Pure NE)**.** *Every finite congestion game has at least one pure Nash equilibrium, and best-response dynamics converge to one in finitely many steps.*

Proof. Immediate from Theorems 7.2, 5.4, and 6.2. □

Remark 7.4. The converse of Rosenthal's theorem also holds: Monderer and Shapley [1996] showed that every finite exact potential game is isomorphic (in best-response structure) to a congestion game. Thus the classes of exact potential games and congestion games essentially coincide.

Example 7.5 (Network Routing)**.** Source s to destination d via two parallel links: $d_A(k) = k$, $d_B(k) = 2 + k/2$, with $n = 4$ users. Each user chooses one link. Profile $(3, 1)$: delays $d_A(3) = 3$, $d_B(1) = 2.5$. No user improves by switching, so this is a NE. The Rosenthal potential selects among multiple NE based on which the dynamics reach.

Example 7.6 (Resource Allocation)**.** n agents choose among m machines with linear congestion $d_e(k) = k\ell_e$. The potential is $\Phi(a) = -\sum_e \ell_e\, n_e(n_e+1)/2$. Pure NE correspond to load-balanced configurations.

8 Connections to Multi-Agent RL

Independent Learners in Potential Games

Potential games guarantee convergence of independent learners: best-response dynamics converge (Theorem 6.2); log-linear learning ($\mathbb{P}(a_i) \propto \exp(\beta\, u_i)$) concentrates on potential maximizers as $\beta \to \infty$; stochastic better-response with exploration converges to the global Φ-maximizer.

Signaling and Emergent Communication

Signaling game theory predicts three outcomes for MARL communication: separating equilibria (meaningful messages), pooling (babbling—a common failure mode), and semi-separating (partial information). The single-crossing property predicts when separating equilibria emerge.

Potential-Based Reward Shaping

The difference reward $D_i(a) = R(a) - R(a_{-i}, c_i)$ satisfies $D_i(a_i, a_{-i}) - D_i(a_i', a_{-i}) = R(a_i, a_{-i}) - R(a_i', a_{-i})$, making R an exact potential for individual payoffs D_i. This turns cooperative MARL into a potential game, ensuring independent learners converge while maximizing the team objective.

Opponent Modeling via PBE

PBE formalizes Bayesian opponent modeling: maintain prior $p(\theta_{-i})$, update via Bayes' rule after observing actions, act optimally under the posterior. The off-path belief freedom warns that learned opponent models may be fragile to surprising actions.

9 Exercises

Exercise 9.1 (Simple Signaling PBE (*)). Consider a signaling game with $\Theta = \{G, B\}$, $\mathbb{P}(\theta = G) = 2/3$, $M = \{m_1, m_2\}$, $A = \{a_1, a_2\}$. Receiver payoffs: $u_R(\theta, m, a_1) = 1$ if $\theta = G$, -1 if $\theta = B$; $u_R(\theta, m, a_2) = 0$. Sender payoffs:

$$u_S(G, m_1, a_1) = 3,\ u_S(G, m_1, a_2) = 1,\ u_S(G, m_2, a_1) = 2,\ u_S(G, m_2, a_2) = 0,$$
$$u_S(B, m_1, a_1) = 1,\ u_S(B, m_1, a_2) = 2,\ u_S(B, m_2, a_1) = 3,\ u_S(B, m_2, a_2) = 0.$$

(a) Does a separating PBE exist where G sends m_1 and B sends m_2?

(b) Does a pooling PBE exist where both types send m_1?

(c) Does a pooling PBE exist where both types send m_2?

Exercise 9.2 (Verifying a Potential Function (**)). Consider the game with $A_1 = A_2 = \{X, Y\}$ and payoffs:

	X	Y
X	$3, 2$	$1, 1$
Y	$2, 1$	$4, 3$

(a) Show this is an exact potential game by constructing Φ.

(b) Find all pure Nash equilibria using Φ.

(c) Trace best-response dynamics from every initial profile and verify convergence.

Exercise 9.3 (Three-Player Congestion Game (**)). Three players share two resources $\{A, B\}$. Each player chooses exactly one resource. Delay functions: $d_A(k) = k$, $d_B(k) = 2k - 1$.

(a) Write the Rosenthal potential Φ.

(b) Enumerate all action profiles (up to symmetry) and compute Φ for each.

(c) Find all pure NE by identifying Φ-maximizers.

(d) Verify your NE by checking no player wants to deviate.

Exercise 9.4 (Spence Signaling with Specific Costs (**)). In Example 4.3, let $H = 2$, $L = 1$, $\pi = 1/2$, $c(H, e) = e/2$, and $c(L, e) = e$.

(a) Find the range of e_H for which a separating PBE exists with $e_L = 0$.

(b) Identify the least-cost separating equilibrium.

(c) Verify the pooling PBE with $e^* = 0$ and compute the pooling wage.

(d) Compare total surplus (worker + firm payoff) in the least-cost separating PBE versus the pooling PBE.

Exercise 9.5 (Not Every Game is a Potential Game (**)). Consider Matching Pennies:

	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

(a) Set up the four equations that an exact potential Φ must satisfy and show they are inconsistent.

(b) Does this game have a pure NE? Relate to Theorem 5.4.

(c) Show that no ordinal potential exists either.

Exercise 9.6 (Network Routing with Three Links (***)). Three parallel links with delays $d_1(k) = k$, $d_2(k) = k + 1$, $d_3(k) = 2k$. There are $n = 6$ users, each choosing one link.

(a) Write the Rosenthal potential function.

(b) Find a pure NE by applying best-response dynamics from the initial profile $(6, 0, 0)$.

(c) Is the NE you found the global maximizer of Φ?

(d) Compute the price of anarchy: the ratio of total delay at the worst NE to the socially optimal total delay.

Exercise 9.7 (PBE with Three Types (***)). Sender has type $\theta \in \{1, 2, 3\}$ with equal probability 1/3. Messages: $M = \{m_1, m_2\}$. Receiver actions: $A = \{a_1, a_2\}$. Payoffs: $u_S(\theta, m, a_1) = \theta$, $u_S(\theta, m, a_2) = 0$; $u_R(\theta, m, a_1) = \theta - 2$, $u_R(\theta, m, a_2) = 0$.

(a) Show that no fully separating PBE exists (three types, two messages).

(b) Find a semi-separating PBE where type 3 sends m_1 and types 1,2 send m_2.

(c) Find a pooling PBE where all types send m_1.

(d) Which equilibrium is most informative? Discuss.

10 Summary

- **PBE** combines sequential rationality with Bayesian belief consistency; every PBE is a BNE (Proposition 3.4).
- **Signaling games:** sender-receiver structure with separating, pooling, and semi-separating equilibria. Spence's model illustrates the single-crossing property. Every finite signaling game has a PBE (Proposition 4.4).
- **Exact potential games:** unilateral payoff changes equal potential changes. Pure NE exist at Φ-maximizers (Theorem 5.4).
- **Finite improvement property:** best/better-response dynamics converge in finite steps (Theorem 6.2).
- **Rosenthal's theorem:** congestion games are exact potential games (Theorem 7.2), hence have pure NE.
- For MARL: potential games guarantee independent learner convergence; difference rewards create potential structure; signaling theory predicts emergent communication outcomes.

11 Further Reading

- **Lesson 37**: Bayesian games and learning dynamics.
- **Lesson 35**: Extensive-form games and SPE.
- Fudenberg and Tirole [1991], Ch. 6, 8: PBE, sequential equilibrium.
- Spence [1973]: original job market signaling paper.
- Monderer and Shapley [1996]: potential games theory.
- Rosenthal [1973]: congestion games.
- Shoham and Leyton-Brown [2008], Ch. 6–7: multi-agent AI connections.

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